Engineering Design as a Research Methodology

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1 When to consider this methodology

A design methodology is normally applied when the research project focuses on the development of a device, process, procedure or system, which could include software. For brevity, in the remainder of this document, "system" will be taken to include all of the other designs. A system is considered to comprise "elements", also sometimes called "subsystems".

The methodology applies to prototype designs, technology demonstrators or concept demonstrators, but can also be applied to feasibility studies and techno-economic analyses that contain a significant design element.

1.1 Design of a prototype

A prototype is a version of a system that is intended to be identical to the first production version of the system. Testing of a prototype may lead to changes to the design, but the changes are unknown at the stage the prototype is made.

For a prototype design project, the research hypothesis could typically be stated as "A design for an xxx (a system) can be created that will fulfil the needs of yyy (one or more end-user or stakeholder)", or the project's research objective could be "design an xxx (a system) to fulfil the needs of yyy (one or more end-user or stakeholder)".

In university research projects of this type, it is often impractical to build and test a prototype as part of the research project. Sometimes only a concept demonstrator (see next subsection) is built and tested. In these situations, care must be taken that the design is not limited to the concept demonstrator (which may be limited to using available components in the lab) but is aimed at the intended eventual application. After the prototype design has been completed, a simplified design can be derived from the prototype design, and the simplified design is then built and tested. This building and testing can be considered an experiment where, as is usual for an experiment, a simplified or idealised scenario is tested to evaluate strategically selected aspects.

1.2 Design of a technology or concept demonstrator

In these projects only selected elements of the system, chosen as part of a bigger strategy, will be designed (and probably built and tested). An example is where, for a design that poses a high technical risk, a project is formulated to develop and test the elements of a system that are exposed to that technical risk, before embarking on prototype development for the system as a whole. Here the research hypothesis can be "xxx (some system elements) of a yyy (a larger system or element) can be created to fulfil zzz (a statement of the performance or goal of xxx)". Alternatively, a research objective can be formulated, for example:

- “Develop and test a concept demonstrator for system yyy”. This formulation is appropriate when the system elements with high technical risks are not known when formulating the research objective.

- "Develop xxx (some system elements) of a yyy (a larger system or element) to fulfil zzz (a statement of the performance or goal of xxx)".

For example, consider a nano-technology-based diagnosing device. A concept demonstrator research project could focus only on the key diagnosing function using laboratory-type sensors and microprocessors, whereas a prototype design would have to use sensors and microprocessors suitable for routine production, and also consider enclosures, power supplies, packing and storage, etc.

1.3 Feasibility studies and techno-economic evaluations

These projects can be encountered when it is uncertain whether a functional design will be possible (requiring a technical feasibility study), what performance reasonably could be expected (requiring a technical evaluation), and/or what the expected costs associated with the system will be (requiring an economic evaluation).

A feasibility study can be aimed at a whole system or only at some of its elements. Therefore, a feasibility study type research project can be similar in this respect to a concept demonstrator design. However, feasibility studies and techno-economic evaluations are often more open-ended, i.e. the study is not aimed at a specific system, but a broader range of similar systems. For example, a narrowly focussed feasibility study could analyse the feasibility (e.g. expressed in terms of performance and costs) of battery-based energy storage systems in a specific site, while a broader study could develop a techno-economic model of the performance and costs of a pneumatic energy storage concept in a range of likely sites.

These research projects share many of the characteristics of a concept design, except that some design targets or constraints will be unknown or broadly defined. Part of the objective in feasibility studies could be to determine a range of feasible values for the design targets, while techno-economic evaluations are often aimed at developing a model (based on first principles, empirical data and simulations like FEA and CFD) that can estimate design targets for a range of constraints.

1.4 Other cases

The principles of the design research methodology are applicable much wider, including even designing experimental setups. In many respects, the design process is a generally applicable problem-solving approach.

2 Reporting sequence

Before considering the typical parts of a design, it is worth noting that the report of a research project should <u>not</u> follow the iterative nature of the design process itself. The report's role is to demonstrate that the result of the research project (the conclusion) is justified and should not be written as a journal of the project.

Experience has shown that when the report of a design-based research project is split into the topics given below and reported in that order, it is the easiest for

the reader to follow. Researchers should resist the temptation when writing up the research to mix the different elements. Even using forward references is preferable to mixing the elements.

The preferred reporting sequence for a relatively simple system is:

a) Describe the problem/opportunity, at least in terms of a set of design requirements or criteria that can be used to judge whether the problem has been solved. If warranted, also document how you arrived at the requirements.

b) Describe the alternative solutions that were considered (the concepts). If warranted, also document how you arrived at the alternative solutions.

c) Evaluate the alternative solutions only in terms of the design requirements. Typically one would first evaluate each concept by itself to eliminate infeasible concepts and then compare all the feasible concepts.

d) Explicitly identify the preferred solution and describe why it was chosen (with reference to the design requirements).

e) If the selected solution contains system elements that justify design refinement, repeat the above steps for each refinement.

When designing more complex systems, the above reporting and its associated design process should be expanded to apply a more complete systems engineering approach, typically starting with developing a system concept definition and a system architecture, followed by designing the system elements.

As implied by the above guidelines, always describe the big picture before the details. The reader will not understand the significance of a system element's design without understanding where that element fits into the system. In system engineering terms: describe the system element's boundary, environment and overall functions before discussing what happens inside the system element.

3 Design requirements

This section outlines only the key features of the development of the design requirements. If the development of the requirements is a major part of the research project, a more complete systems engineering approach would be appropriate. For more information about this approach, please refer to the presentation "Systems Engineering Highlights for Final Year Projects", the Systems Engineering Templates of the Department of Mechanical and Mechatronic Engineering and the lectures in the module Introductory Systems Engineering 444.

3.1 Terminology

The development of the design requirements is also called a "system concept definition" (in systems engineering contexts) or a "requirements analysis". The terms "needs analysis" and "specification development" are also used, but these terms can also refer to a specific part of the requirements analysis. The outputs

of this process are sometimes called "engineering requirements" or "engineering specifications".

3.2 Role

A design project can only be successfully completed if the design requirements are defined clearly and sufficiently complete. Without a proper description of the design requirements, it will be impossible at the end of the project to conclude whether the design that was produced, is successful. However, it is often impossible in a research project to state the design requirements fully at the start of a research project (e.g. in feasibility studies or techno-economic analyses). This sets a design-based research project apart from a normal design project in industry, where the design requirements are frozen early in the project. A design-based research project should normally produce new knowledge, not only a new design for a system.

The nature and scope of the design requirements will differ for projects of different types, which are outlined in Section 1. If a prototype design project includes building and testing, the scope of what is built and tested may be reduced (e.g. to a technology demonstrator or concept demonstrator), but usually, the design requirements should not be restricted to that limited scope from the outset. The report on the testing phase can rather start with a revised design, aimed specifically at the test model, with some additional requirements (e.g. it must fit into a specific test environment) and relaxing of some previous requirements. Also, in feasibility studies or techno-economic analyses, some design requirements may be unknown or only broadly specified at the outset. However, even then all the known significant requirements should be documented to the extent possible.

3.3 Project schedule considerations

The development of the requirements should start early in the project based on, e.g., literature, user surveys or studying competing designs' functions and performance specifications. However, the refinement of the requirements will typically continue through the concept development phase and even into the detail design phase, as the researcher learns more about the system being designed. On the other hand, researchers should minimise reworking previously completed parts of the design and should, therefore, consider whether revising previous design requirement targets will be worthwhile. In some cases, a change with a large impact should be recorded in the report as possible further work.

If a researcher knows little about the system when starting out, it may be better to build concept demonstrators early in the design process. The process of designing, building and testing the concept demonstrators can provide valuable insight into the design itself and, thereby, into the design requirements.

3.4 Typical aspects included in the design requirements

The following elements should be considered for inclusion in the requirements analysis, but note that the extent of the consideration of each element will depend on the particular project.

- Identify the stakeholders to be considered in the design. This should normally include the stakeholders from the whole system life cycle and can, therefore, include end-users, manufacturers, maintainers, recycling concerns, etc.
- Distinguish between and describe the boundary between the wider system of interest (WSoI) and the narrower system of interest (NSoI). The WSoI contains all elements that will have a significant effect on the design, while the NSoI contains only the system elements that can be determined (modified, designed) by the designer. The NSoI is part of the WSoI. For example, if the NSoI is a car, the WSoI can include the road, weather, other traffic, refuelling stations and even the factory that makes the car. Limit this discussion to what has a significant influence on the design project because there is no limit to what can be considered to have an influence on the NSoI.
- Describe the information, energy and/or material flows that cross the boundary between the WSoI and the NSoI, as well as spatial/geometric constraints. Describe the elements in the WSoI, outside the NSoI, that interact with these flows but avoid stating the obvious (which will be seen as waffle). Assign measurable parameters to these flows. If the values of these parameters are determined outside the NSoI, the parameters are called design-independent parameters (DIPs) or design constraints. If the designers of the NSoI have the authority to determine the values of the parameters, they are called technical performance measures (TPMs), which are further described below.

  The DIPs document what the NSoI must be able to contend with (e.g. ambient temperatures) or what the NSoI can expect to be provided to it. For example, for electric power supplied from outside the NSoI, DIPs could include the voltage range (220 to 250 V single phase AC) and allowed power range (10 kW max). For another example, consider a system that heats water drawn from the WSoI where a DIP could be “Water inlet temperature” with a range of (4 to 30 °C).
- Describe the functional requirements (FRs), i.e. the main functions the NSoI must perform. Note that a "function" describes an action that a stakeholder wants the system to perform, ideally without prescribing (or even implying) what technology, device or system element will perform the function. In other words, describe "what" the system must do without saying "how" it will do it. For example, a function could be "heat water" and should not be "water heater".

  In functional analyses for the system as a whole, it is important to avoid referring to system elements inside the NSoI because the selection of those elements should be part of the concept development phase. Also, it is prudent to distinguish between primary functions (i.e. functions that the external stakeholders are directly concerned with) and derived functions (i.e. functions that are required by the primary functions). When

documenting the requirements of a system, it is best to avoid over-refinement of derived functions.

Often in research projects, only the functions relevant to the system's operation need to be considered, but it would add theoretical depth and practical value if other life cycle phases are included. In some research projects, the production/construction, support/maintenance and/or retirement phases must be included to be able to fairly compare the eventual designs with competing systems.

Various diagrams can be considered to document a functional analysis, but in most reports, only one of the types should be used. Functional flow block diagrams, as used in Systems Engineering, should only be considered for complex systems. For simpler software, flow charts may be appropriate to describe the functions. For systems where interactions with system elements outside the NSoI are important (as in many mechatronic systems), interaction diagrams (also known as "swimlane" diagrams) are useful. Another format to consider is a state diagram if a state machine is a useful metaphor to describe the behaviour of the NSoI.

- Describe the non-functional requirements (NFRs) or technical performance measures (TPMs). TPMs are parameters that can be used to determine how well the NSoI performs the required functions. TPMs can be quantitative or qualitative and, if possible, the TPMs' target ranges must be specified. However, in some research projects (such as concept evaluations and feasibility studies), typical or feasible target values are only determined during the research. For the above-mentioned example of a "heat water" function, the associated TPMs can be "Water outlet temperature" with a range of "95 to 100 °C" and "Water outlet flow rate" with a range of "0 to 5 L/s". The target ranges assigned to the TPMs must ensure the NSoI will meet the stakeholders' needs (in other words, they will be used in later design phases to determine how well a concept meets the stakeholders' expectations).

  Ideally, the TPMs should be independent of one another (i.e. changing one TPM should not necessarily require a change in another TPM). In a systems engineering approach, each TPM is assigned to at most one function, while a function may have multiple TPMs associated with it. However, there should be no contradictory requirements (TPMs and DIPs). For example, if a system is required to heat water, the TPMs specifying the outlet water temperature and the outlet flow rate must not be in conflict with the DIPs that specify the inlet water temperature and the maximum electrical power that may be drawn.

- "Industry best practice" should not be itemised in the requirements, but can be assumed to be complied with. Application of this guideline requires some judgement and it may be best to err on the side of specifying too much, rather than too little. However, the overall project objective should be taken into account: the requirements definition must be sufficient for

the research objectives to be met, but not necessarily to withstand the scrutiny of a professional enquiry.

In a research document, all of the above information must be justified. Never use "magic numbers", i.e. numbers that just appear from nowhere. Some parameters, e.g. TPM target values, may be left undetermined initially if the research's objective is to determine the technical or techno-economic feasibility of the NSoI. Otherwise, cite a reputable source (published reference) or describe the work done to determine the information (e.g. interview, analysis, evaluation of similar systems, etc).

4 Concept development

4.1 Role

Unless the research objective is restricted to evaluating a particular concept, the generation of concepts should be described. In the broader research context, "concepts" can be alternative solutions, approaches and, for techno-economic analyses, even analysis models that can be considered.

The objective of this section is to convince the examiners that the candidate has done what was reasonably possible to counter "conceptual vulnerability". This implies that the report must convince the examiners that the process used to generate and select concepts avoided:

- That the best concept (or near-best) was not considered.
- That the best concept was inadvertently eliminated when choosing the preferred concept.

4.2 Concept generation

In most academic research, innovation is not an evaluation criterion, but the application of scientific methods is. The emphasis in the report should, therefore, be on the method that was used to generate concepts, as well as the resulting concepts. Typical methods to consider include literature and patent searches, first principles arguments (such as degrees of freedom that must be constrained or left unconstrained, systematic consideration of load paths), idealising the concept to fundamental mechanisms (e.g. cams, cranks, fulcrums, wedges, etc), considering different actuation methods or energy sources, etc.

If the system’s complexity justifies it, the functional analysis in the design requirements may be refined before generating concepts for each main function. For simpler systems, a morphological chart can be used. For more complex systems, a systems engineering approach that progressively refines the system definition can be used. In the latter case, the top-level system elements are first chosen and then each of those elements is refined until a solution can be selected.

4.3 Concept descriptions

The report must describe the overall concepts and, if appropriate, sub-concepts in sufficient detail so that the examiners can understand the concept evaluation and selection. Please refer to Section 4.6 when deciding whether to place a description in the main body of the report or an appendix.

The chosen interactions between system elements may be a significant part of a concept's definition. To document those interactions, UML's interaction (or swimlane) diagrams or sequence diagrams can be considered. If sequences are not important, a diagram with the system elements and interconnecting lines that represent the flow of information, material and energy, can be considered. For detailed mechanical and electro-mechanical designs, a simplified 3D model of the concept can be shown with, if relevant, the actuators (motors, cylinders, etc.) and power transmission elements (belts, gears, shafts, etc). It is important that such 3D models are approximately to scale because spatial relationships are often important when evaluating the feasibility of concepts.

4.4 Concept evaluation and selection

In a research report, the following is important:

- Only use design requirements previously identified in the report to evaluate and compare concepts. Often, previously unidentified requirements become apparent when concepts are evaluated and then the design requirements must be revised accordingly.
- The selection of concepts must be justified using scientific or engineering arguments. Here also, beware of magic numbers (as described above). Every value or metric used for concept selection must be justified.

In the research report, guard against mixing a description of the concept and the evaluation of the concept. Always first describe the concept and then evaluate the concept.

A concept that cannot reasonably be expected to meet an important design requirement (i.e. the concept is not viable), can be described, evaluated and immediately eliminated. It adds no value to compare infeasible concepts with other concepts. Therefore, weeding out unviable concepts is often the first step in concept selection.

Avoid quantitative methods (e.g. numerical performance scores) for concepts that are broadly described with too little detail to justify numerical performance scores. Rather use qualitative arguments for concepts that are too vague to analyse. Some qualitative arguments may be quantitative in nature, but with only broad estimates available at this stage. For example, manufacturing and assembly costs could be roughly estimated using mass and parts count. Even though these parameters can only be quantified precisely after a detailed design has been done, a qualitative argument could compare unrefined concepts on the basis of "higher" or "lower" part counts.

If quantitative methods, like QFD or a Pugh matrix, are used:

- Each entry in the tables must be justified (no magic numbers).
- Use scoring numbers that are normalised, but weight the different requirements (again, no magic numbers: the weights must be justified).
- The scoring "resolution" must be in proportion to the accuracy of the estimate of the extent to which the concept meets a requirement (e.g. do not use a 10-point scale if your estimates are not near 10% certain).
- If the weighted sum of concepts' scores is used, note that this assumes that the "performance model" is linear, while often that would be an oversimplification.
- The conclusions drawn after adding up weighted scores must take into account the uncertainty in the allocated scores. The final scores will also have an uncertainty band and differences within that band should be considered to be insignificant. Do a sanity check by varying the allocated scores within their uncertainty bands for a few highly weighted requirements and check whether the ranking of final scores changes.
- Keep in mind that a quantitative method is not necessarily objective. Acknowledge subjectivity in the report. The influence of subjectivity can be reduced by combining multiple, independent subjective opinions when that is feasible.

4.5 Concept refinement and re-evaluation

In all situations, it is important that when comparing concepts, all concepts must be developed to a similar level of detail. It may be much easier to see problems with a concept developed to high detail, while bigger problems with a concept that has only been broadly defined, may still be hidden.

If many concepts are considered, it is usually not worth the effort to refine all of them to the level where detailed (e.g. quantitative) evaluation is reasonable. Therefore, it is often advisable to use qualitative arguments to reduce the concepts to a small number (e.g. two or three) and then refine the shortlisted concepts before comparing them further.

4.6 Main body or appendix

The main body of the report must always describe the selected concept and give an overview of the motivation for choosing that concept, so that the readers can get an impression of the process used.

Unless the concept development and selection are central parts of the research report, many of the details of these processes will only be given in appendices. If readers are concerned with the detail (such as when they disagree with the selected concept or the science of the process applied), they can refer to the appendices.

5 Detail design and analysis

As indicated in Section 4.5, the refinement process can already start during concept development. This section considers the (further) refinement after the preferred concept has been chosen.

5.1 General guidelines

This part of the research is highly dependent on the particular project, but the following guidelines should still be considered.

- Detail design involves many choices at progressively increasing levels of detail. Each of these choices can be seen (and documented) as a small design project in its own right. For each element of the design, keep to the recommended order in the report:
  - First describe the requirements (or objectives) for the system element being designed;
  - Then describe the alternatives that were considered;
  - Often this is followed by some analyses and evaluation (note that this part may overlap with the design evaluation described in the next section, but detailed information should not be repeated in the report); and
  - Finally, explicitly state the alternative that was chosen and explain why that is a good choice.
- The level of detail of the design process for each system element must be appropriate for the level of the system element. For minor design details, a sentence or two, or a short paragraph, addressing all the steps of the design process may be sufficient. The examiners just need to be convinced that the appropriate alternative solutions were considered and that the choice of the final solution is scientifically justified (except as pointed out below).
- Sometimes a scientific argument cannot, within the time and resource constraints of the research project, be given why one alternative is better than another. In such a situation, be honest in the report by, for example, stating "since the circumstances in which the research was done did not allow for a sufficient comparison of options A and B, option A was chosen as a subjective preference, but option B should be considered in future research."
- It should be clear to the readers how the design requirements for a system element were derived from the requirements for the system as a whole. It is not necessary to "state the obvious" in the report but, in this respect, err on the side of saying too much rather than too little.

5.2 Cost analyses

Analysing (or even just predicting) system cost can be very challenging and researchers must take care to acknowledge the limitations of their cost analyses.

It is usually very important to categorise the costs according to the system life cycle stages, e.g. research cost, design and development cost, manufacturing or production cost, purchase cost, operating cost, maintenance or support costs, and retirement costs. Purchase costs can contain many elements, e.g. factory gate costs (manufacturing and overhead costs, which often include the design cost), wholesale markup, retail markup, marketing costs and commissions, product liability insurance costs, provisions for product callbacks, etc. A mistake some novice designers make is to compare their estimate of a prototype's production cost with the retail cost of a competing system, while for some systems, the wholesale price is double the factory gate price and the retail price is double the wholesale price.

Note that environmental impact can also be considered to be a "cost".

6 Design evaluation

The success of the design produced in the research project must be determined in relation to the design requirements that are documented earlier in the report. Note that the design evaluation may overlap with the analyses and evaluation mentioned in concept development and detail design, but there the focus is usually on evaluating an element of the system, while here the focus is on evaluating the system as a whole.

The Glossary of the System Engineering Body of Knowledge gives the following definitions of terms that should be used when documenting the design evaluation:

- Verification: Confirmation, through the provision of objective evidence, that specified (system) requirements have been fulfilled.
- Validation: Confirmation, through the provision of objective evidence, that the (stakeholder) requirements for a specific intended use or application have been fulfilled.

The term "evaluation" is not defined in the Glossary. In research contexts, it usually implies subjectivity in an assessment. Showing that a system element meets its FRs and NFRs can be considered to be verification, but showing that the system as a whole meets the overall FRs and NFRs (which express the stakeholders' requirements) can be considered to be validation. Discussing whether the research project's objectives were met, will be an evaluation.

The basis for claims that the design meets a requirement may be a combination of a first principles analysis[1], an empirical analysis (i.e. using previously published experimental results), a numerical analysis (e.g. FEA or CFD) or testing. In some cases, a design review by experts can be used as a design evaluation. The authority of the experts must be justified if this route is used.

Testing should be conducted and reported as a scientific experiment. Similarly, simulations using tools such as CFD or FEA should be presented as a form of

[1] Avoid the use of "theoretical analysis" since this term does not convey clearly what the basis of an analysis is.

testing and reported as an experiment. These simulations will, in some reports, be part of the detailed design (e.g. when used to choose between alternatives) and in others part of the design evaluation (when only applied to the final choices).

The differences between prototypes (Section 1.1) and technology or concept demonstrators (Section 1.2) should be kept in mind when planning the testing and reporting on it.

Evaluating the results of a feasibility study is usually similar to the evaluation of a concept demonstrator, but evaluating a techno-economic analysis is more challenging and usually involves comparison with published cases and/or application of the techno-economic model to a case study (please refer to the lecture notes on "Case Study as an Engineering Research Method").

7 Conclusions

These notes give numerous guidelines for research projects where there is a substantial design element. It cannot be overemphasised that the content and scope of the documentation of the design approach must be tailored to the specific project. In some projects, some aspects mentioned above will be trivial and hardly mentioned in the report. In other projects, more expansive processes and documentation (e.g. using a systems engineering process) would be appropriate.

Always keep in mind that the report of the research project must be aimed at providing evidence that supports the conclusions at the end of the report and that the conclusion must indicate whether the research question, hypothesis or objective has been answered or achieved. The design approach described above is a good, general way to provide such evidence.

In the report's conclusion, be honest about the limitations of a design. A superficial claim in the report's conclusion that the "design works" or only stating its benefits and not also its shortcomings casts doubt on the science underlying the research and the objectivity of the researcher. A research project should attempt to <u>prove and disprove</u> the research hypothesis, before concluding whether the hypothesis is true. Beware of claiming that a design is novel, the best, the first in the country, the cheapest, etc. because it is very difficult to prove such a claim. In a research report, all claims must be substantiated.